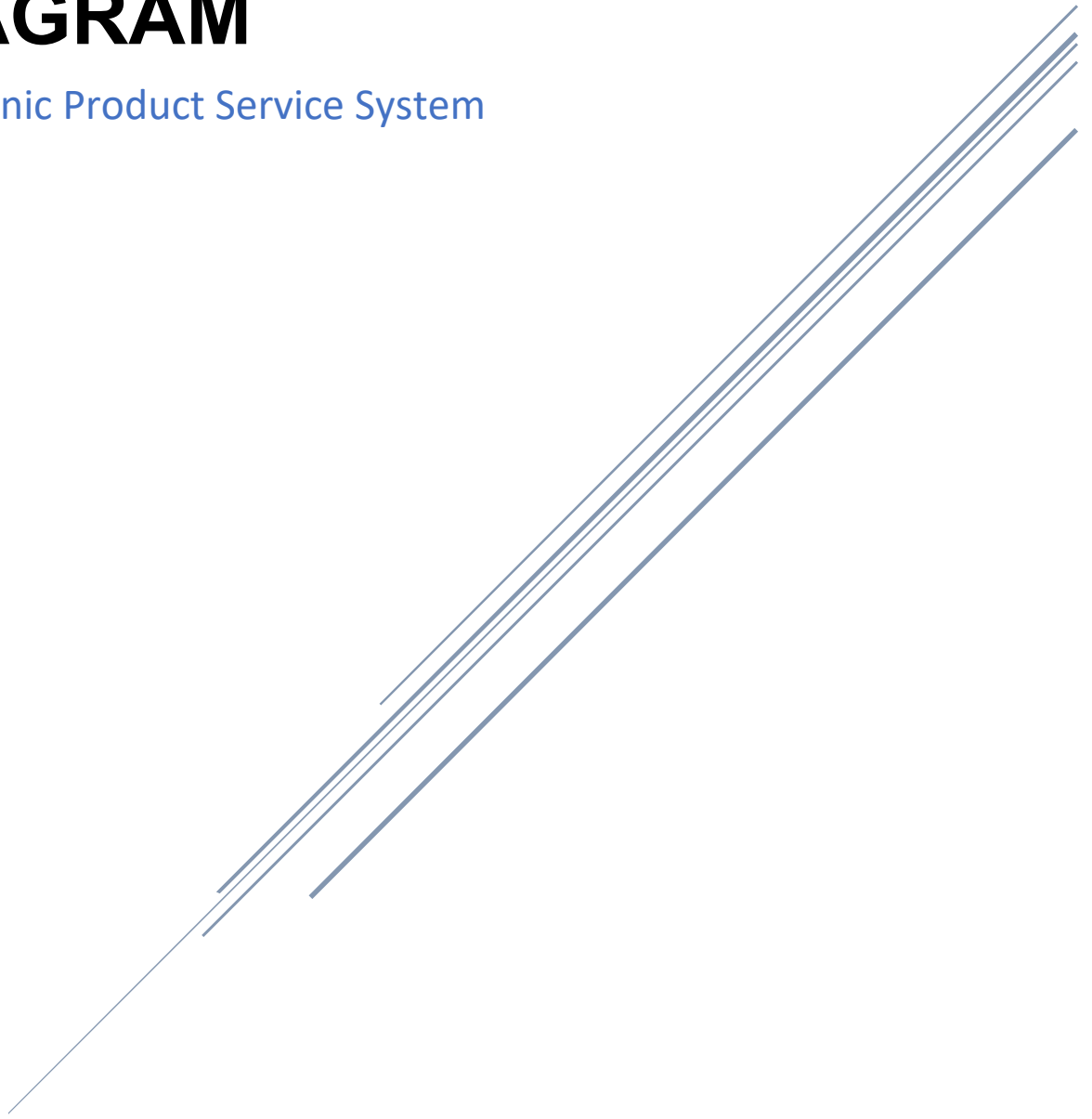


ENTITY RELATIONSHIP DIAGRAM

Electronic Product Service System



SEMESTER PROJECT : ERD

BS-IT -IV(M)

COURSE CODE:CC-215

DATABASE SYSTEMS

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2023-2027

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PROBLEM STATEMENT

Electronic Product Service System:

This system supports sales of electronic products like laptops and smartphones. Products come with basic or extended warranties. When a customer initiates a service request, the system logs the device, issue, technician assigned, repair history, and whether replacement parts were used. It also flags serial numbers of products that frequently fail for quality control. The system integrates manufacturer recall data to auto-notify customers affected.

Method:

The method for making Entity Relationship Diagram for statement problem is as following:

- ✚ Read statement carefully
- ✚ Identify Entities
- ✚ Identify Attributes
- ✚ Identify Relationships
- ✚ Check Cardinality
- ✚ Make Diagram

Core Entities

Strong entities are:

- Customer
- Product
- Technician
- Manufacturer

Weak entities are:

- Warranty
- ServiceRequest
- RepairHistory
- ReplacementPartUsed

Attributes

1. Product

- ProductID (primary key)
- ProductType (e.g., laptop, smartphone)
- ModelNumber

- SerialNumber
- Description
- ReleaseDate
- Price

2. Customer

- CustomerID (primary key)
- Name
- Email
- Phone
- Address
- City
- State
- Country
- PostalCode

3. Technician

- TechnicianID (primary key)
- Name
- Expertise
- Experience
- Certification
- ContactInformation

4. Manufacturer

- ManufacturerID (primary key)
- Name
- Address
- City
- State
- Country
- ContactInformation

1. Warranty

(weak entity, dependent on Product)

- WarrantyID (partial key)
- ProductID (foreign key, part of composite key)
- WarrantyType (basic or extended)
- ExpirationDate
- CoverageDetails
- TermsAndConditions

2. ServiceRequest

(weak entity, dependent on Product and Customer)

- ServiceRequestID (partial key)
- ProductID (foreign key, part of composite key)
- CustomerID (foreign key, part of composite key)
- IssueDescription
- Status
- RequestDate
- ResolutionDate

3. Repair History

(weak entity, dependent on ServiceRequest)

- RepairHistoryID (partial key)
- ServiceRequestID (foreign key, part of composite key)
- RepairDate
- Description
- Cost
- TechnicianID (foreign key)

4. ReplacementPartUsed

(weak entity, dependent on RepairHistory)

- ReplacementPartUsedID (partial key)
- RepairHistoryID (foreign key, part of composite key)
- ReplacementPartID (foreign key)
- Quantity

Relationships

1. **Product** has **Warranty** (one-to-one)
2. **Customer** makes **ServiceRequest** (one-to-many)
3. **ServiceRequest** is for **Product** (many-to-one)
4. **ServiceRequest** is assigned to **Technician** (many-to-one)
5. **ServiceRequest** has **RepairHistory** (one-to-many)
6. **RepairHistory** uses **ReplacementPartUsed** (many-to-many)
7. **Product** is manufactured by **Manufacturer** (many-to-one)

Critical Relationship

- **Purchase(1:N)** Customer purchases product

- **Has**(1:1)Mandatory warranty link
- **Generates**(1:N)Service request to repair history
- **Triggers**(M:N)Recall service request association

Cardinality

1. A **product** can have one **warranty**.
2. A **customer** can make many **service requests**.
3. A **service request** is for one **product**.
4. A **service request** is assigned to one **technician**.
5. A **service request** can have many **repair histories**.
6. A **repair history** can use many **replacement parts**.

Assumptions

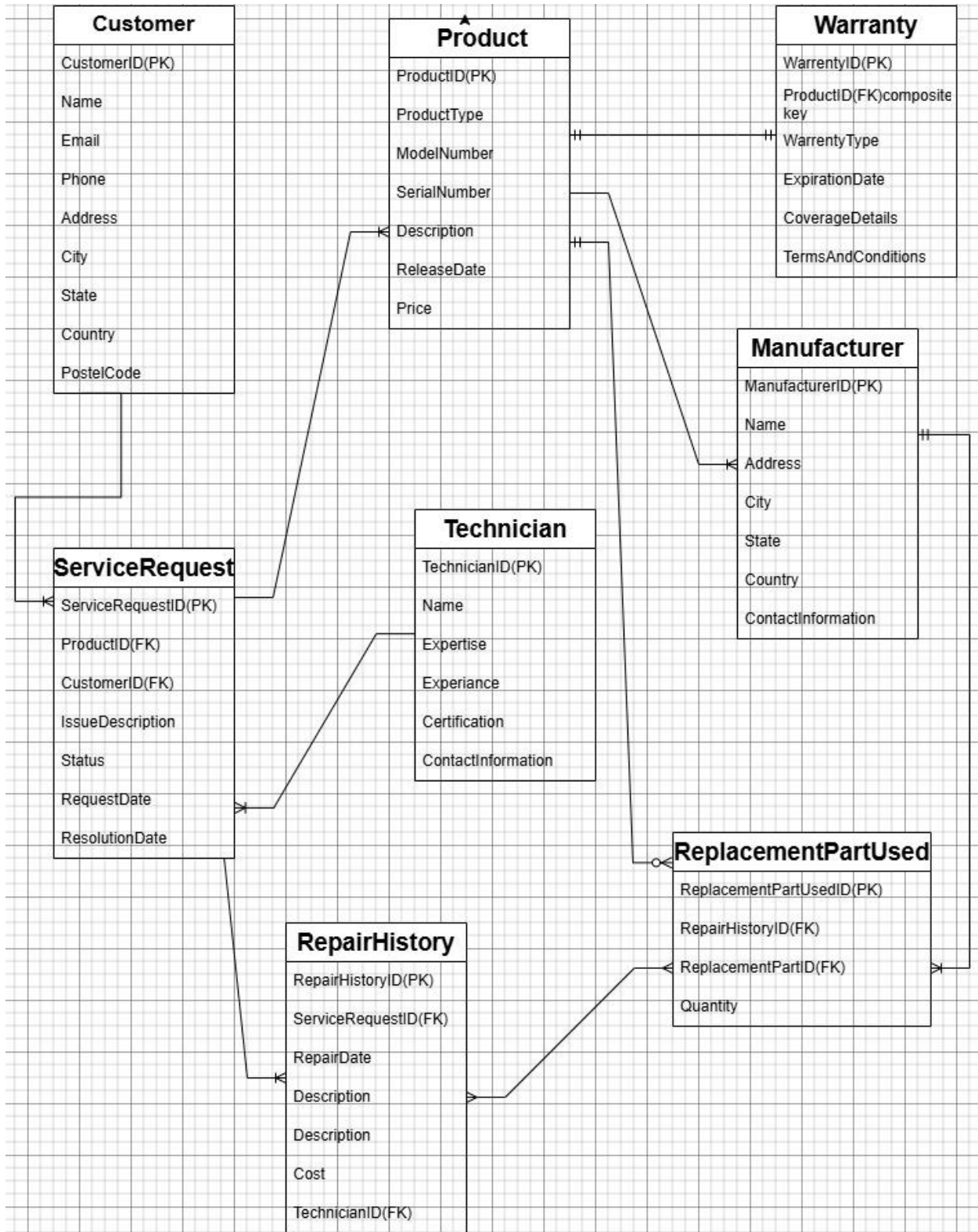
1. Each **product** has a unique serial number.
2. Each **customer** has a unique email address.
3. Each **technician** has a unique name and expertise.
4. Each **replacement part** has a unique part number.

Diagram



ERD.drawio

Crow's Foot Model



Chen Model

